

A Multi-Agent Perception-Action Alliance for Efficient Long Video Reasoning

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Abstract

This paper presents a multi-agent perception-action exploration alliance, dubbed A4VL, for efficient long-video reasoning. A4VL operates in a multi-round perception-action exploration loop with a selection of VLM agents. In each round, the team of agents performs video question-answer (VideoQA) via perception exploration followed by action exploration. During perception exploration, each agent learns to extract query-specific perception clue(s) from a few sampled frames and performs clue-based alignment to find the video block(s) that are most relevant to the query-specific event. During action exploration, A4VL performs video reasoning in three steps: (1) each agent produces its initial answer with rational, (2) all agents collaboratively scores one another through cross-reviews and relevance ranking, and (3) based on whether a satisfactory consensus is reached, the decision is made either to start a new round of perception-action deliberation by pruning (e.g., filtering out the lowest performing agent) and re-staging (e.g., new-clue and matching block based perception-action exploration), or to conclude by producing its final answer. The integration of the multi-agent alliance through multi-round perception-action exploration, coupled with event-driven partitioning and cue-guided block alignment, enables A4VL to effectively scale to real world long videos while preserving high quality video reasoning. Evaluation Results on five popular VideoQA benchmarks show that A4VL outperforms 18 existing representative VLMs and 11 recent methods optimized for long-video reasoning, while achieving significantly lower inference latency. Our code is released at <https://github.com/git-disl/A4VL>.

1. Introduction

Multimodal large language models (MLLMs) have been developed to support video understanding in recent years.

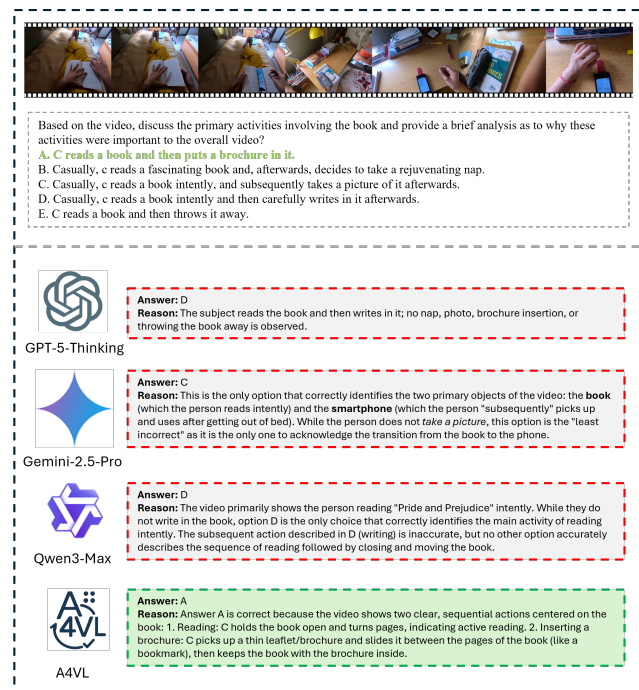


Figure 1. Responses from A4VL and three of the strongest closed-sourced models.

Compared to traditional approaches [11, 14, 46, 56, 57, 59], MLLM-based methods [15–17, 33, 37, 38, 41, 49, 51, 53, 54] exhibit strong ability for spatiotemporal visual reasoning, as well as human-level text understanding and generation. However, most models often fail to produce high performance reasoning on long videos, because when the number of frames is large, the memory and time costs scale quadratically. Even with enough computing resources and time, some recent works [20, 29] show that simply increasing the frame sampling density may even hurt performance, for example, due to redundant frames introducing noise and distracting attention away from truly informative key

frames. The problem is especially aggravated for video QA workloads that center on questioning events covered in only a small subset of frames, as the problem of identifying accurately where the most relevant frames reside is challenging. Efficient and scalable solutions are on-demand to improve video understanding performance for complex vision–language reasoning about long and complex videos.

Our work is inspired by some pioneering efforts in this direction. For example, GPT-4o [25] is trained on a large multimodal corpus and, after such training, is able to process long videos by taking many more frames. Although this yields higher accuracy, it is much slower. On Video-MME [7], GPT-4o takes more than 150 s on average to answer a multiple-choice question. Therefore, the community has begun to explore optimization strategies. Some works focus on token merging [1, 55], sparsification [9, 13, 18, 20], or training-free agent approaches [2, 10, 35, 40, 45, 47, 52], while others design more efficient neural architectures [8, 44, 50] or memory-retrieval-based systems [22, 31, 34, 48]. For example, DYT0 [55] proposes a dynamic bipartite token merging strategy, which supports a higher frame sampling rate while keeping the resource usage acceptable; AuroraLong [44] replaces the LLM component of MLLMs with a linear RNN layer to handle input sequences and also combines token merging techniques to improve efficiency. Among these methods, agent-based approaches and memory retrieval methods are orthogonal to token-based and architecture-focused optimizations. In this paper, we present an innovative multi-Agent Perception-Action Alliance based method for long video reasoning.

Problem Statement. We note some limitations of existing agent-based methods. First, their inference speed is low. For example, VideoAgent [5] takes over ten minutes on a long video that lasts for an hour. Second, most of them use a single MLLM to perform decision making and inference; they do not support effective collaboration among multiple agents. In addition, they often rely on strong video grounding models, and yet when the question is long or complex, they remain struggling to locate key frames. For example, MoReVQA [24] uses object detection and image QA models to find keyframes and achieves good performance on NeXT-QA [43]. However, its accuracy falls behind other approaches on the long and complex video datasets, e.g., EgoSchema [23] (see Table 1 in Section 3.2).

To address these limitations, we propose a multi-agent perception–action alliance framework (A4VL), featured with three core components: (i) an agent teaming strategy that leverages multiple diverse MLLMs to enhance video reasoning performance; (ii) an event-based partitioning and random sampling strategy to handle resource-intensive long-video processing; and (iii) an iterative agent coordination framework for perception exploration and action exploration that further boosts reasoning performance. Figure 1

shows the responses from three of the strongest closed-sourced models to date (GPT-5-Thinking [26], Gemini-2.5-Pro [4], and Qwen3-Max [39]), as well as our A4VL, on an example from the EgoSchema dataset. Although all these MLLM-based methods generate incorrect answers, A4VL remains correct. Figure 2 shows the general framework of A4VL. Figure 3 illustrates our agent teaming strategy, which selects the team that collaborates best from a pool of agents. Figure 4 in Section 2 visualizes the reasoning procedure of A4VL on this example.

Upon receiving a query, A4VL first goes through the perception exploration step. During this step, each agent samples a small number of frames (e.g., 4) to generate a perception clue. We then apply an event-based partition method to divide the video into several blocks and use models like CLIP [30] to obtain the similarity score of each block, measuring its consistency with the perception clue. Each agent samples from high-scoring blocks and forms a set of frames for the next step, e.g., 16 sampled frames. Next, agents move into the action exploration step: they first generate answers and reasons for the sampled frames relevant to a query. If the answers are consistent, a summarizer aggregates the reasons and returns the final answer to the user. Otherwise, each agent is asked to rate every other agent, and the agent team is pruned based on these scores. The surviving agents then revise their perception clues based on the current state and move back to Stage 2 of the perception exploration step, starting the next round of collaboration.

Our main contributions are summarized as follows:

- **A4VL framework.** We develop A4VL, a training-free multi-agent perception–action framework for long-video question answering under tight frame budgets.
- **Teaming and clue-guided perception.** We empower A4VL with an unsupervised agent teaming method and an event-based *sample*→*clue*→*block* pipeline, optimized by combining CLIP-similarity enhanced block selection with adaptive frame allocation, enabling A4VL agents to focus on key frames without retraining.
- **Accuracy–latency gains.** On five long-video benchmarks against 28 baselines, A4VL achieves the best accuracy while being substantially faster than strong closed- and open-source MLLMs that process many more frames.

2. Methodology

2.1. Design Overview

In this section, we provide an architecture overview of A4VL. As shown in Figure 2, A4VL consists of a perception exploration step and an action exploration step, where multiple agents work collaboratively to produce the final answer. The two steps run iteratively until agents reach a consensus. When we have multiple MLLMs available, we prefer to use a small subset as the agent pool. Hence, we de-

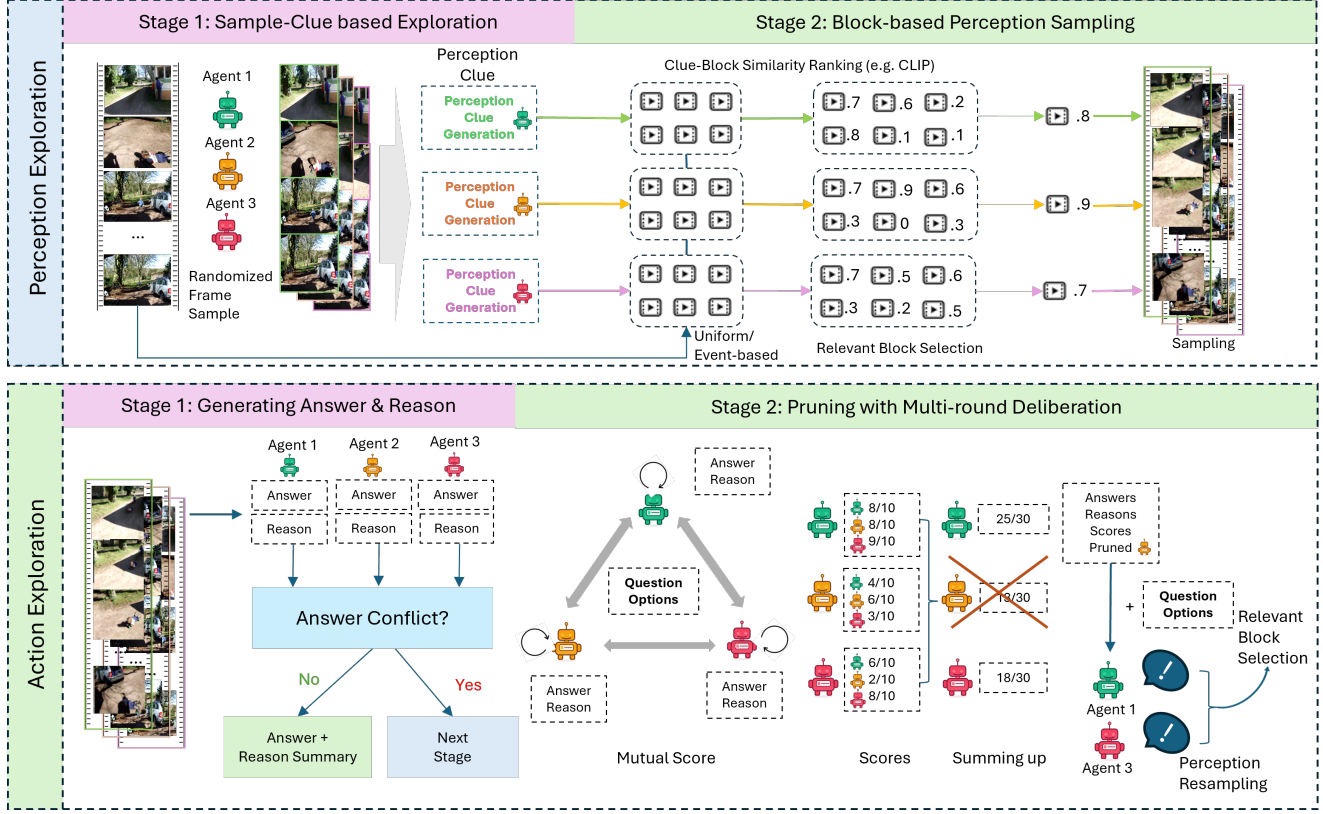


Figure 2. Overview of A4VL architecture.

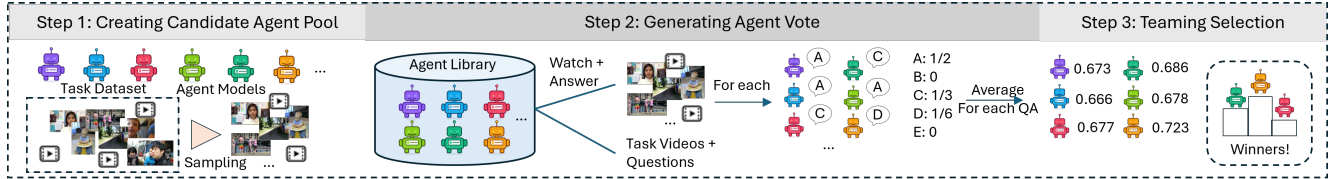


Figure 3. A4VL agent teaming workflow.

sign an optional agent teaming stage (Figure 3) that selects a task-specific subset of agents. Thereafter, all questions under the same task reuse the same agent subset. The agent teaming process is as follows. Suppose we have M agents and K unlabeled video-question pairs from the task, each with C answer choices. These K samples can be randomly drawn from the task dataset. Using labels would make the teaming algorithm more accurate, but we assume unlabeled data for practical deployment. In the agent teaming step, each agent independently runs the perception exploration and answer generation steps to produce predictions. We count the frequency of each choice in each sample and denote them as $\{f_{qr}\}_{q=1:K, r=1:C}$, where each f_{qr} is the fraction of agents that choose the r th option for question q . For each agent, let r_q denote the index of the option it selects for question q . We define its score as $\frac{1}{K} \sum_{q=1}^K f_{qr_q}$. We

choose m ($m = 3$ in our implementation) agents with highest scores to form the task-specific agent pool. As illustrated in Figure 3, the teaming process has three steps.

- **Step 1:** we prepare the agent library and sample a small set of video-question pairs from the task.
- **Step 2:** each agent runs part of the A4VL workflow and independently generates answers, and we compute the frequency of each answer for each question. For the example in Figure 3, the frequencies of choices A–E are 1/2, 0, 1/3, 1/6, and 0, respectively. Purple, blue, and light green agents therefore receive a score of 0.5 on that question, while the green and pink agents receive 1/3, and so on.
- **Step 3:** we average each agent’s score over all sampled questions, and the top 3 agents (green, yellow, and pink) are selected for this task.

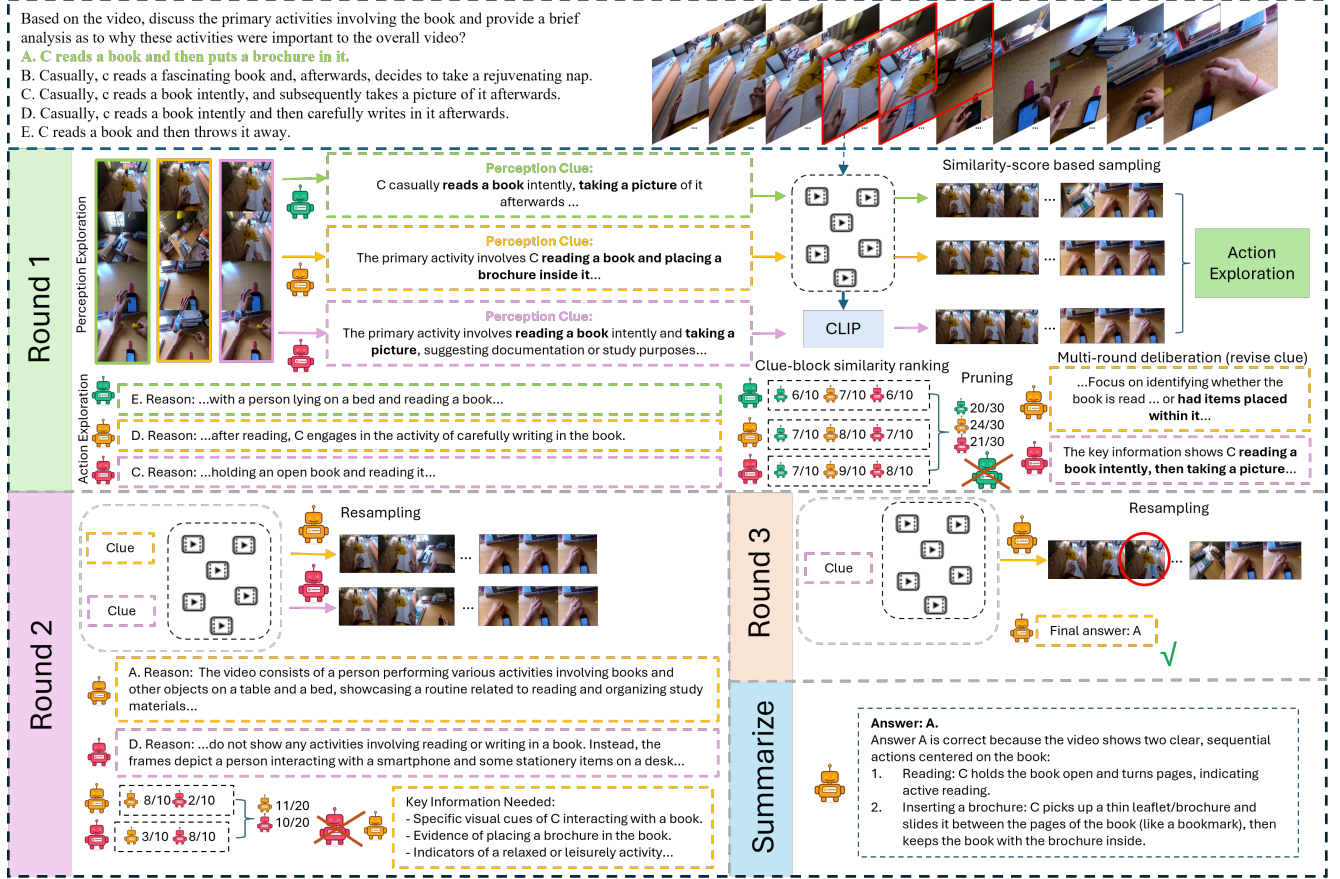


Figure 4. Example of A4VL on EgoSchema. Here all agents have wrong answers at the beginning, but later one of them turns to the correct answer and other agents are pruned.

Within the task, after A4VL receives a video and the user’s question (including choices, and additional information like subtitles under some benchmarks), it first goes through the perception exploration step. As shown in Figure 2, the perception exploration step consists of two stages. In Stage 1, each agent randomly samples N_1 frames ($N_1 = 4$ in our settings) to preview the entire video, and then generates a perception clue to identify keyframes. In the illustrative example, the video is divided into six blocks using an event-based approach. For each agent, a CLIP model like [3] is used to compute similarity scores between its perception clue and each block. These scores indicate how relevant each block is to the current question. Then, N_2 frames (set to 16 in our experiments) are sampled from the blocks according to their scores, as described in the perception exploration subsection, and are used in the next action exploration step.

During the action exploration step, each agent is asked to provide an answer and a corresponding reason, given its N_2 frames obtained from the previous step. If they reach a consensus, then we give the answer to the user and summa-

rize the reason to explain the answer. Otherwise, each agent is asked to evaluate all agents’ answers, and the agent with the lowest total score (based on self- and peer evaluations) is pruned. For the example shown in Figure 2, the yellow agent is pruned and other agents are required to generate a new perception clue in order to better distinguish between conflicting answers in the next round. After that, all agents will go back to the stage 2 of perception exploration to start a new round, until finally all agents reach a consensus.

2.2. Perception Exploration

The top part of Figure 2 shows our perception exploration step workflow. We index rounds by j . In the first round, each agent goes through the entire perception exploration step; in subsequent rounds ($j > 1$), each agent starts directly from selecting relevant blocks. Suppose we have a video V which consists of n frames: $V = \{v_1, v_2, \dots, v_n\}$, question Q , options O , and agent pool $\mathbb{A} = \{A_i : 1 \leq i \leq m\}$, where m is the number of selected agents after agent teaming. As shown in Figure 2, perception exploration step has two stages: sample-clue based exploration and block-

based perception sampling.

In the first stage, each agent generates a perception clue, which contains necessary information that can be used to identify keyframes in stage 2. For each agent $A_i \in \mathbb{A}$, we first sample N_1 frames from the video using a sampling strategy $p_1 : 2^V \times \mathbb{N}^+ \rightarrow 2^V$:

$$\hat{v}_{A_i} = p_1(V, N_1). \quad (1)$$

For p_1 , we have two choices: (i) we can randomly select N_1 frames from the entire video, or (ii) we can divide the video into event-based blocks, and sample a few from each block. In experiments, we test both alternatives and find that (i) is better, so we use choice (i) as our final choice. The reason is that agents do not need fine-grained understanding of the video to generate a coarse perception clue, so uniform random sampling provides higher diversity and better temporal coverage. Given the sampled N_1 frames, question, and options, each agent is asked to generate a perception clue, which is later used to identify relevant blocks. We denote the perception clue for agent A_i at j th round as $P_{i,j}$. In the first round,

$$P_{i,1} = A_{i,clue}(\hat{v}_{A_i}, Q, O). \quad (2)$$

For subsequent rounds, $P_{i,j}$ is generated at the end of the action exploration step. For example, by the end of the first round, $P_{i,2}$ is generated for each $A_i \in \mathbb{A}$.

In the second stage, we first partition the video into at most B blocks, $B_1, B_2, \dots, B_b$. We provide two partition strategies: (i) **uniform partition**, which means $B_i = V_{(i-1)\lfloor \frac{n}{B} \rfloor + 1 : i\lfloor \frac{n}{B} \rfloor}$; (ii) **event-based division**, where we detect scene changes using DINOv2 embeddings [27] and simple pixel cues (HSV/motion/sharpness), generate candidates with KTS [28], PELT [12], and SSM-based novelty [6], merge them and apply non-maximum suppression (NMS), and keep the top $B - 1$ boundaries (at most B blocks). This procedure is efficient, as most videos in our benchmarks can be processed within two seconds. Details are given in the appendix. We also test both strategies in our experiments, and finally choose (ii). After dividing the video into blocks, we compute the similarity between each block and each agent's perception clue. In this paper, we use a CLIP model to compute the similarity scores. Other approaches, such as computing the similarity between the caption of each block and each perception clue, are also possible, but since CLIP is the fastest, we leave the exploration of more advanced similarity computation as future work. We then sample the target frames under two scenarios. (i) If all blocks have similarity scores below ρ (set to 0.8 in our settings), we sample from the most relevant block for each agent: for agent A_i , we identify the block $B_{i,j}^*$ with the maximum score and use a sampling strategy

$p_2 : 2^V \times \mathbb{N}^+ \rightarrow 2^V$ to sample N_2 frames from $B_{i,j}^*$:

$$V_{act}^{(i)} = p_2(B_{i,j}^*, N_2). \quad (3)$$

(ii) Otherwise, we retain blocks with similarity score $> \rho$. For each agent A_i , we obtain a score vector $\mathbf{s}^{(i)}$ over blocks. We first normalize it by $\mathbf{s}^{(i)} \leftarrow \mathbf{s}^{(i)} - \max(\mathbf{s}^{(i)})$, and then compute an integer allocation vector $\mathbf{c}^{(i)} := \lfloor N_2 \cdot \text{SoftMax}(\mathbf{s}^{(i)}) \rfloor$ for the blocks. If $\sum_k c_k^{(i)} < N_2$, we randomly distribute the remaining $N_2 - \sum_k c_k^{(i)}$ samples over the blocks to ensure a total of N_2 action frames. We then sample $c_k^{(i)}$ frames from each retained block B_k using p_2 and take their union to form $V_{act}^{(i)}$.

2.3. Action Exploration

Now each agent has its own set of N_2 sampled frames, denoted by $V_{act}^{(i)}$. The action exploration step uses these frames to generate answers and, when necessary, to prune agents. It has two stages: answer and reason generation, and pruning with multi-round deliberation, as shown in the bottom part of Figure 2.

In the first stage, each agent is asked to generate an answer and corresponding reason given its own sampled frames and the question information. For agent A_i , it generates the answer $a_{i,j}$ and reason $R_{i,j}$ as follows:

$$a_{i,j} = A_{i,act}(V_{act}^{(i)}, Q, O), \quad (4)$$

$$R_{i,j} = A_{i,reason}(V_{act}^{(i)}, a_{i,j}, Q). \quad (5)$$

Then for all agents, we have the answer set $S_{a,j}$ and reason set $S_{r,j}$:

$$S_{a,j} = \{a_{1,j}, a_{2,j}, \dots, a_{|\mathbb{A}|,j}\}, \quad (6)$$

$$S_{r,j} = \{R_{1,j}, R_{2,j}, \dots, R_{|\mathbb{A}|,j}\}. \quad (7)$$

If all agents reach a consensus, we send all $P_{i,j}$ values in round j , $S_{a,j}$, and $S_{r,j}$ to our reason summarizer to give the user the final answer and an explanation. We consider two ways to check for consensus: (i) **Majority consensus**: the most frequent answer appears more than $|\mathbb{A}|/2$ times; (ii) **Full consensus**: all agents output the same answer. We test both strategies and ultimately adopt (ii).

If the agents do not reach a consensus, then we step into the second stage. Given $S_{a,j}$, $S_{r,j}$, Q and O , each agent evaluates the soundness of its answer and every other agent's answer. It is asked to give a rating from 1 to 10:

$$\{s_{i,\mathbb{A}_1}, \dots, s_{i,\mathbb{A}_{|\mathbb{A}|}}\} \leftarrow A_{i,eval}(Q, S_{a,j}, S_{r,j}), \forall A_i \in \mathbb{A}. \quad (8)$$

In Figure 2, the yellow agent has a rating of 8/10, 6/10 and 2/10 given by three agents. After that, for each agent, we

sum up the ratings given by all agents:

$$s_{A_i} := \sum_{A_k \in \mathbb{A}} s_{k, A_i} \quad (9)$$

For example, the yellow agent now has an overall score of 16/30, which is the lowest. Suppose it's A_2 , then we have $s_{A_2} = 16$. Now we do agent pruning, where the agent with the lowest score is removed from the agent pool:

$$\mathbb{A} \leftarrow \mathbb{A} \setminus \{A_{min}\}, \quad (10)$$

$$A_{min} := \arg \min_{A \in \mathbb{A}} s_A. \quad (11)$$

In the Figure 2 case, $A_{min} = A_2$, and the yellow agent is removed. For each of the remaining agent A_i , we generate $P_{i,j+1}$ using $P_{i,j}, S_{a,j}, S_{r,j}, A_{min}, Q, O$ to generate the refined perception clue, which is used in the next round:

$$P_{i,j+1} = A_{i,refine}(P_{i,j}, S_{a,j}, S_{r,j}, A_{min}, Q, O). \quad (12)$$

Figure 4 shows an example from EgoSchema. We can see at the beginning, all agents generate different and incorrect answers. However, in the second round, the orange agent begins to realize that choices B, C, D, and E do not have clear visual evidence, and it revises the perception clue to focus on the interaction between the person and the book. In the third round, the keyframe where the person is putting a brochure into the book is sampled (highlighted by the red circle), and therefore the orange agent continues to generate the correct answer. Finally, our summarizer reads the entire multi-round process and returns the final answer together with a concise explanation to the user. The pseudocode of A4VL is provided in the appendix.

3. Experiment

3.1. Experimental Settings

We evaluate A4VL on five benchmarks that cover both short and long videos, with an emphasis on long-video reasoning.

To test our method on short videos with strong temporal/causal reasoning requirements, we evaluate A4VL on NeXT-QA [43]. We randomly sample 200 QA pairs per question type from its test split, resulting in a subset of 1,493 QA pairs. EgoSchema [23] is a medium-length video QA dataset consisting of videos lasting about 3 minutes. We evaluate on its test split with ground-truth labels, which contains 500 QA pairs.

For long videos, we evaluate on LongVideoBench [42], MLVU [58] (test split), and Video-MME [7]. LongVideoBench consists of 1,337 QA pairs, and each video is accompanied by subtitles; many questions are strongly subtitle-related. The videos range from about 3 minutes to 2 hours in length. MLVU-Test is the harder test split of MLVU, containing roughly 500 videos with

durations between 3 minutes and 2 hours, and the task is to select the correct answer out of six options. Video-MME consists of 2,700 QA pairs and is divided into short, medium, and long groups according to video duration, with 900 QA pairs in each group. Videos in the short group last about 10 seconds, whereas videos in the long group can last for over an hour.

In the agent teaming step, we select $m = 3$ agents from a pool of 8 models: LLaVA-Video-7B-Qwen2 [54], QwenVL-2.5-7B [38], InternVL3.5-8B [41], QwenVL-2.5-32B [38], InternVL3.5-38B [41], InternVL3-78B [41], QwenVL-2.5-72B [38], and LLaVA-Video-72B-Qwen2 [54]. Selecting $m > 3$ agents may further boost the accuracy score, but it will harm the efficiency. Unless otherwise specified, all experiments involving A4VL are conducted on six H200 GPUs.

3.2. Results

We compare A4VL against two closed-source MLLMs, GPT-4o [25] and Gemini 1.5 Pro [37], 16 open-source MLLMs, and 10 agent-based or long-video-oriented methods. As shown in Table 1, A4VL outperforms all baselines on every benchmark. On NeXT-QA, accuracy saturates near 80% as the backbone size increases; the best baseline, InternVL3-78B, reaches 84.0%, while A4VL further improves to 85.1%. On EgoSchema, InternVL3-78B achieves 76.8%, and A4VL is the only method exceeding 80%, with 82.2% (+3.2). LongVideoBench is where we observe the largest gain: compared to GPT-4o at 66.7%, A4VL (built solely from open-source models) reaches 72.2% (+5.5). MLVU-Test is substantially harder, with more answer choices and more complex question types (e.g., tracking specific players in long sports videos), yet A4VL still improves over prior methods by 1.9–19.6 points. On Video-MME, A4VL surpasses all baselines across all duration groups and on the overall average. Notably, although the strongest single MLLM backbone reaches at most 67.1% without subtitles, A4VL attains 77.2% on average, 2.2 points higher than the best closed-source model, Gemini 1.5 Pro (75.0%). The models selected by our agent teaming step for each benchmark are listed in Table 2.

Table 3 reports the average inference time per sample for four representative methods—the closed-source MLLM GPT-4o, the strong open-source MLLM InternVL3-78B, and two well-known agent-based systems, VideoAgent and TravelER—as well as our A4VL. We measure efficiency on NeXT-QA, EgoSchema, and MLVU, which correspond to short-, medium-, and long-video settings. On short videos (NeXT-QA), all methods except TravelER exhibit similar and acceptable latency. As video length increases, however, the inference time of other methods grows rapidly; for example, on MLVU, even GPT-4o takes over two minutes to answer a single question, whereas A4VL scales

Table 1. A4VL compared with other methods on five benchmarks. For Video-MME, the result is in format w. subtitle/wo. subtitle. For Video-MME, the improvement row shows the improvement without subtitle, since A4VL is mainly designed to optimize visual reasoning. * near the model names mean we take the results from the model paper or benchmark leaderboard.

Model	NeXT-QA	EgoSchema	LongVideoBench	MLVU	Video-MME			
					Short	Medium	Long	Average
GPT-4o* [25]	-	72.2	66.7	54.9	80.0/82.8	70.3/76.6	65.3/72.1	71.9/77.2
Gemini 1.5Pro* [37]	-	71.1	64.0	-	81.7/84.5	74.3/81.0	67.4/77.4	75.0/81.3
LLaVA-Video-7B-Qwen2 [54]	81.2	61.8	61.1	51.7	72.4/75.6	58.1/63.3	50.7/60.4	60.4/66.4
VideoLlama3-7B [49]	67.4	52.6	52.5	43.9	80.1/80.2	63.7/69.6	54.9/61.0	66.2/70.3
InternVL3.5-8B [41]	76.2	62.0	53.6	47.1	69.6/74.0	56.3/63.9	47.7/61.9	57.9/66.6
LLaVA-NeXT-Video-7B [53]	39.7	39.6	43.5	26.3	36.4/40.1	30.8/38.0	27.9/17.2	31.7/31.8
QwenVL-2.5-7B [38]	76.8	60.7	54.7	47.7	72.1/73.6	55.7/63.9	48.2/60.9	58.7/66.1
LongVA-7B* [51]	69.3	-	51.3	41.1	61.1/61.6	50.4/53.6	46.2/47.6	52.6/54.3
LongVU-7B* [33]	-	37.6	-	-	-	-	-59.5	-160.6
VideoChat2-7B* [17]	61.7	56.7	39.3	30.1	48.3/52.8	37.0/39.4	33.2/39.2	39.5/43.8
LLaVA-Video-72B-Qwen2 [54]	83.7	69.4	60.7	51.5	77.8/80.0	64.9/70.8	58.6/73.0	67.1/74.6
InternVL3.5-38B [41]	78.9	75.4	57.0	56.1	76.1/79.1	58.6/68.9	57.4/68.3	64.0/72.1
InternVL3-78B [41]	84.0	76.8	56.4	55.3	78.6/80.4	65.4/73.4	56.7/67.8	66.9/73.9
LLaVA-NeXT-Video-34B [53]	70.1	46.6	50.5	39.5	63.1/66.4	51.1/53.2	44.6/48.7	52.5/56.0
QwenVL-2.5-32B [38]	76.8	64.8	53.3	48.5	73.2/77.1	61.6/68.2	54.1/68.6	63.0/71.3
QwenVL-2.5-72B [38]	79.6	70.7	56.1	53.7	75.1/77.0	66.3/73.3	59.6/70.8	67.0/73.7
LLaVA-OneVision-72B* [15]	80.2	62.0	63.2	-	76.7/79.3	62.2/66.9	66.0/62.4	66.3/69.6
Aria-25.3B* [16]	-	-	63.0	-	67.9/78.3	67.0/71.7	58.8/66.3	67.6/72.1
ViperGPT [36]	56.9	24.8	-	24.9	-	-	-	-
SeViLA [19]	64.0	28.4	-	31.7	-	-	-	-
VideoAgent [5]	66.1	55.0	-	34.3	-	-	-	-
TravelER [32]	66.0	55.4	-	36.3	-	-	-	-
MoReVQA* [24]	69.2	51.7	-	-	-	-	-	-
VideoRAG-72B* [21]	-	-	65.4	-	81.1/-	72.9/-	73.1/-	75.7/-
AuroraLong* [44]	-	-	-	52.7	-	-	-	-
DYTO-34B* [55]	72.9	56.8	-	-	-	-	-	-53.4
BOLT* [20]	79.5	64.0	59.6	-	70.1/-	66.0/-	49.6/-	59.9/-
DynFocus* [8]	-	-	31.8	49.6	50.9/53.9	43.7/46.0	37.7/43.6	44.1/47.8
LVAgent [2]	83.0	78.4	66.9	50.0	82.4/83.6	72.8/78.6	66.3/76.4	73.9/79.5
A4VL	85.1	82.2	72.2	58.0	86.6/87.3	76.8/83.2	68.3/77.9	77.2/82.8
Improvement$\uparrow$	1.9 $\sim$ 45.4	3.8 $\sim$ 57.4	5.3 $\sim$ 41.6	1.9 $\sim$ 19.6	4.2 $\sim$ 20.2	2.5 $\sim$ 46.0	0.9 $\sim$ 40.4	2.2 $\sim$ 45.5

Table 2. Models used for each benchmark. Intern-78B refers to InternVL3-78B [41], Intern-38B refers to InternVL3.5-38B [41], QwenVL-72B refers to QwenVL-2.5-72B [38], and LLaVA-72B refers to LLaVA-Video-72B-Qwen2 [54].

Benchmark	Models		
NeXT-QA	Intern-78B	Intern-38B	QwenVL-72B
EgoSchema	Intern-78B	Intern-38B	QwenVL-72B
LongVideoBench	Intern-78B	Intern-38B	QwenVL-72B
MLVU	Intern-78B	Intern-38B	LLaVA-72B
Video-MME	Intern-78B	LLaVA-72B	QwenVL-72B

Table 3. Average inference time per sample on four representative methods and A4VL.

Model	NeXT-QA	EgoSchema	MLVU
GPT-4o	23s	54s	127s
InternVL3-78B	15s	50s	204s
VideoAgent	20s	83s	175s
TravelER	101s	94s	450s
A4VL	18s	37s	74s

much more gracefully. For TravelER, the inference time on EgoSchema is lower because its official implementation uses image captions only to extract visual information on

this dataset, yet A4VL still requires about 60% less time than TravelER.

3.3. Ablation Study

Since multi-round collaboration is a core component of A4VL, we first study how the maximum number of rounds affects accuracy. In our default implementation, A4VL can run at most three rounds because there are three agents in total. In this ablation, we stop the procedure after a fixed maximum round and take the majority answer at that round as the final prediction; if there are multiple majorities, we choose the answer with the highest score. Figure 5 shows how accuracy changes as the maximum number of rounds increases. On all benchmarks, accuracy consistently improves when more rounds are allowed. Table 4 reports the actual number of rounds used in the default A4VL setting. As the dataset becomes harder, agents tend to collaborate for more rounds. Even within Video-MME, when subtitles are available the task becomes easier and thus requires fewer rounds.

We also test several design choices discussed in Section 2, using EgoSchema as the testbed. We first vary the sampling strategies. For perception-frame sampling p_1 , we

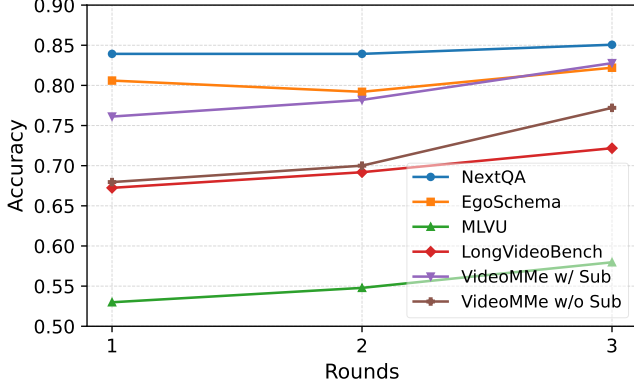


Figure 5. Relationship between accuracy and maximum number of rounds.

Table 4. Number of actual running rounds in A4VL. The data of Video-MME is shown in w/o subtitle / w. subtitle format.

#Rounds	1	2	3
NeXT-QA	1072	230	191
EgoSchema	303	101	96
LongVideoBench	566	341	430
MLVU	183	116	203
Video-MME	1289/1667	557/421	854/612

either perform random sampling over the entire video (R) or sample one frame from each event-based block (E). For action-frame sampling p_2 , we either sample from uniformly divided blocks (R) or from event-based blocks (E). We evaluate three variants: RRSampling, RESampling, and ERSampling, where the first character denotes p_1 and the second denotes p_2 . A4VL corresponds to RESampling. The accuracy and latency of each variant are shown in Table 5. Among all strategies, RESampling achieves the best accuracy with similar latency, so we adopt it as our default. Intuitively, perception exploration benefits from coarse, uniformly distributed frames to cover the whole video, while action exploration benefits from frames concentrated on active events. From the table, we can also know the event-based partitioning only takes about 2 seconds.

Table 5. Accuracy and latency of different sampling strategies on EgoSchema. R means random sampling and E means sampling from event-based blocks. The first character represents the sampling strategy for N_1 perception frames, and the second character represents the sampling strategy for N_2 action frames.

Method	RRSampling	RESampling	ERSampling
Accuracy (%)	80.2	82.2	79.6
Time/Sample	35s	37s	37s

Next, we compare different consensus criteria. Table 6 shows results on EgoSchema under majority consensus (MConsens) and full consensus (FConsens), as defined in

Section 2.3. Both standards yield strong accuracy; FConsens has an accuracy of 82.2%, which is better than MConsensus (81.4%) but incurs higher (still moderate) latency. The reason is FConsens benefits from increasing collaboration rounds to make sure the final answer is strongly confident. Since the additional latency is acceptable, our default method uses FConsens.

Table 6. Results of different consensus conditions on EgoSchema. M means majority and F means full.

Method	MConsens	FConsens
Accuracy (%)	81.4	82.2
Time/Sample	26s	37s

Finally, we test whether agent pruning is necessary. We design two variants without pruning. **NoPruneSum** selects the answer with the highest total score, and **NoPruneMaj** selects the majority answer as the final prediction (breaking ties by total score). For both variants, the early stop condition is either (i) all agents in one round generates the same answer (FConsens) or (ii) in two subsequent rounds, the highest score corresponds to the same answer. As shown in Table 7, no matter how we choose the final answer, removing pruning leads to lower accuracy and substantially higher latency, indicating that pruning is important for both effectiveness and efficiency.

Table 7. Results of different pruning strategies on EgoSchema.

Method	NoPruneSum	NoPruneMaj	A4VL
Accuracy (%)	80.8	79.4	82.2
Time/Sample	60s	60s	37s

4. Conclusion

We presented A4VL, a training-free multi-agent framework that tightly couples perception exploration with action exploration for long video question answering. By combining event-aware block partitioning, clue-guided frame selection, and multi-round agent alliance with consensus and pruning, A4VL turns a pool of heterogeneous MLLMs into a coordinated reasoner over long videos under fixed compute budgets. Across five different benchmarks, A4VL consistently improves over strong baselines and prior agent-based systems, and our round-level ablations show stable gains from each collaboration round.

Looking ahead, we believe that perception-action coordination at the agent level is a promising direction for long-video understanding. Extending A4VL to audio-text-video tri-modal inputs, richer task-conditioned similarity functions, and neuro-symbolic techniques are natural next steps for enhanced perception exploration toward high performance general long-horizon video agents.

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